TORUS-GRID VIETORIS–RIPS COMPLEXES AT HALF THE DIAMETER

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Abstract. For every integer $k \geq 3$, we determine the homotopy type of the closed Vietoris–Rips complex of the $2k$ by $2k$ periodic Manhattan grid at radius k. It is $S^3 \vee \bigvee^{4k} S^{2k-1}$, as conjectured by Adams, Adetowubo, Barriga-Acosta, Feng and Sterling. Adjacent pairs of rows and columns give the high-dimensional spheres. Removing one facet from each leaves a complex with the integral homology of a three-sphere. The comparison is proved uniformly through a homotopy calculation for finite cylinders and a cover by coordinate semicircles.

1. Introduction

Vietoris–Rips complexes of periodic samples provide finite models in which one can follow the change from local geometry to higher-dimensional topology. For a torus grid with the periodic Manhattan metric, Adams, Adetowubo, Barriga-Acosta, Feng and Sterling [1] determined several families of homotopy types. Their Theorem 5.8 identifies the small-scale torus regime, while Theorems 5.10 and 5.12 describe the first two transition families. Their Question 8.6 asks for the homotopy type at half the diameter of an even square grid. We answer that question for all $k \geq 3$; the case $k = 2$ was already known.

For $n \geq 1$, write

$$T_{n,n} = (\mathbb{Z}/n\mathbb{Z})^2, \qquad \delta_n(a, c) = \min\{|a - c|, n - |a - c|\},$$

using representatives $0, \ldots, n - 1$, and set

$$d((a, b), (c, e)) = \delta_n(a, c) + \delta_n(b, e).$$

The closed complex $\mathrm{VR}(T_{n,n}; r)$ contains a finite subset precisely when every pair in it has distance at most r. It is the clique complex of its graph of edges.

Theorem 1. *For every integer $k \geq 3$,*

$$\mathrm{VR}(T_{2k,2k}; k) \simeq S^3 \vee \bigvee_{j=1}^{4k} S^{2k-1}.$$

The metric diameter here is $2k$, so the radius is exactly half the diameter. The wedge is taken at a common base point. For $k = 2$ the same formula gives nine three-spheres, agreeing with the known homotopy type in [1].

The $4k$ high-dimensional spheres have concrete representatives: each pair of adjacent rows or columns induces a cross-polytope boundary. We remove one suitable facet from each of these spheres. Their remaining union has a good cover whose nerve is the join of two circles. To compare this union with the rest of the Rips complex, we use coordinate semicircles. The main calculation is a cylinder whose circular direction has circumference $2k$; adding a column adds exactly one $(2k - 1)$-sphere. This also explains why the equality at the closed threshold matters.

Throughout the proof, fix $k \geq 3$, put $n = 2k$, and write $K = \mathrm{VR}(T_{n,n}; k)$. All homology groups have integer coefficients, and simplicial complexes are identified with their geometric realizations when discussing homotopy.

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2. Adjacent rows and columns

For $b \in \mathbb{Z}/n\mathbb{Z}$, let B_b be the induced subcomplex of K on the two rows with second coordinates $b, b+1$. For $a \in \mathbb{Z}/n\mathbb{Z}$, let D_a be the induced subcomplex on columns $a, a+1$. All indices in these definitions are taken modulo n.

Lemma 2. *Each B_b and D_a is the boundary of an n-dimensional cross-polytope. In particular, it is homeomorphic to S^{n-1}. The faces*

$$\tau_b = \{(a,b) : a \text{ even}\} \cup \{(a+k, b+1) : a \text{ odd}\},$$

and their coordinate-swapped counterparts υ_a in D_a, are maximal faces of K. Each belongs to only its own band among the $2n$ complexes B_b, D_a.

Proof. Every pair within one row is adjacent. Between the two rows the only nonedges are

$$(a,b), \quad (a+k, b+1).$$

These form n opposite pairs, so the clique complex is the cross-polytope boundary. The selected face chooses one vertex from each pair.

To check maximality in K, consider a vertex (x,y) outside the two rows. Its circular distance from at least one of them is at least two. The first coordinates selected in either row form a parity class; their maximum circular distance from x is at least $k-1$. Thus a vertex of the selected face is at distance at least $k+1$ from (x,y). Inside the band, maximality follows from the opposite-pair description.

The two occupied rows identify the horizontal band uniquely. Its first-coordinate projection contains a parity class of $k \geq 3$ points, so it cannot be contained in two adjacent columns. The vertical claims follow by interchanging the coordinates. □

Let

$$L = \bigcup_b B_b \ \cup \ \bigcup_a D_a.$$

Delete the $2n$ maximal faces τ_b, υ_a, retaining all their proper faces, from K and L. Denote the resulting subcomplexes by K^- and L^-, respectively.

Lemma 3. *There is a homotopy equivalence*

$$K \simeq K^- \vee \bigvee^{2n} S^{n-1}.$$

Proof. In each band sphere, the complement of the interior of its selected facet is a closed ball. By Lemma 2, this ball contains no other deleted facet. It therefore lies in K^-, and contracts the attaching map of the selected facet's boundary. Restoring the facets one at a time gives the asserted wedge. Equivalently, apply the null-homotopic union lemma from [1, Lemma 5.7]: if both inclusions from $X \cap Y$ into X, Y are null-homotopic, then

$$X \cup Y \simeq X \vee Y \vee \Sigma(X \cap Y).$$

Here Y is the restored simplex and $X \cap Y$ its boundary. The complexes remain connected because their one-skeletons are unchanged. □

Proposition 4. *The complex L^- is homotopy equivalent to S^3.*

Proof. Cover L^- by the $2n$ band spheres with their selected facets removed. Each member is a closed ball. Two horizontal bands intersect when their indices are adjacent, in a full row simplex; three distinct horizontal bands have empty intersection. The same holds vertically. A mixed intersection is a simplex on a subset of a two-by-two grid, whose pair distances are at most two. None of these intersections contains a deleted facet.

Every nonempty intersection is therefore contractible. The nerve of this cover is $C_n * C_n$, where C_n is the simplicial cycle with n vertices: the horizontal and vertical intersection conditions are independent. The finite subcomplex nerve theorem [2, Theorem 10.6(i)] gives

$$L^- \simeq C_n * C_n \simeq S^1 * S^1 \cong S^3.$$

□

3. Coordinate restrictions

We will compare K^- and L^- on coordinate semicircles. A cyclic interval of length ℓ means $\{c, c+1, \dots, c+\ell\}$, with all entries reduced modulo n.

Lemma 5. *Every simplex of K has at least one coordinate projection contained in a cyclic interval of length k.*

Proof. Embed the grid in $\mathbb{R}^2 \oplus \mathbb{R}^2$ by

$$v(a,b) = (\cos(\pi a/k), \sin(\pi a/k), \cos(\pi b/k), \sin(\pi b/k)).$$

For angular coordinate distances $\alpha, \beta \in [0, \pi]$, the pair inner product is

$$\cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}.$$

It is nonnegative exactly when $\alpha + \beta \le \pi$. The cases where the second factor is zero also have $\alpha + \beta = \pi$. Thus clique vertices have pairwise nonnegative inner products.

Write them as (x_i, y_i). If the convex hull of the x_i excludes zero, choose its closest point p to zero. Minimality along the segment from p to x_i gives $p \cdot x_i \ge |p|^2 > 0$; the first coordinates lie in an open semicircle. Otherwise choose nonnegative λ_i, summing to one, with $\sum_i \lambda_i x_i = 0$, and put $w = \sum_i \lambda_i y_i$. Since $|(x_i, y_i)|^2 = 2$,

$$|w|^2 = \Big|\sum_i \lambda_i (x_i, y_i)\Big|^2 \ge 2\sum_i \lambda_i^2 > 0.$$

Moreover $y_j \cdot w \ge 0$ for every j, by the pair inner products. The second coordinates consequently lie in a closed semicircle. For integer coordinates, either semicircle can be enlarged to one of the stated intervals. □

Finite cylinders. For $1 \le m \le k+1$, let F_m be the closed Rips complex at radius k on $\{0, \dots, m-1\} \times \mathbb{Z}/n\mathbb{Z}$, with metric

$$|x - x'| + \delta_n(y, y').$$

Only the first coordinate is unwrapped.

Proposition 6. *For $1 \le m \le k+1$,*

$$F_m \simeq \bigvee^{m-1} S^{n-1},$$

where a zero-fold wedge is a point. Deleting the selected facets of its $m-1$ adjacent-column bands leaves a contractible complex F_m^-.

Proof. The case $m = 1$ is a simplex. For $m \ge 2$, put $t = m - 1$, and let

$$U = \bigcup_{c \in \mathbb{Z}/n\mathbb{Z}} \mathrm{st}_{F_m}(t, c), \qquad E = F_{m-1} \cap U.$$

Here the closed star consists of faces that can be extended by the indicated vertex. The vertices of the last column form a simplex. All nonempty intersections of their closed stars are cones at any

of the corresponding last-column vertices. Their nerve is a full simplex, so U is contractible, and $F_m = F_{m-1} \cup U$.

Cover E by $E_c = F_{m-1} \cap \mathrm{st}_{F_m}(t, c)$. For each nonempty proper set $S \subset \mathbb{Z}/n\mathbb{Z}$, there is a z with

$$(*) \qquad z \notin S + k, \qquad \delta_n(z, s_0) \le 1 \quad \text{for some } s_0 \in S.$$

If some $s \in S$ has $s + k \notin S$, take $z = s$. Otherwise $S = S + k$, and a cyclic edge from S to its complement gives s_0, z as required.

The vertex $(t-1, z)$ belongs to every E_c, $c \in S$, since $\delta_n(z, c) \le k - 1$. If (x, y) belongs to their intersection, put $r = t - x$. Then $r + \delta_n(y, s_0) \le k$, and

$$d((t-1, z), (x, y)) \le r - 1 + \delta_n(z, s_0) + \delta_n(s_0, y) \le k.$$

Thus the intersection is a cone at $(t-1, z)$. The intersection over all n centers is empty: the second-coordinate antipode of any previous-column vertex gives distance $r + k > k$. The nerve is therefore the boundary of an $(n-1)$-simplex, and $E \simeq S^{n-2}$.

Inductively F_{m-1} is $(n-2)$-connected. This connectivity follows from its wedge description and cellular approximation [3, Theorem 4.8]. Both maps from E to F_{m-1} and U are null-homotopic. The null-homotopic union lemma gives

$$F_m \simeq F_{m-1} \vee \Sigma E \simeq \bigvee^{m-1} S^{n-1}.$$

The facet-removal argument of Lemma 3 also applies inside this cylinder:

$$F_m \simeq F_m^- \vee \bigvee^{m-1} S^{n-1}.$$

It follows that F_m^- has trivial reduced integral homology. Its fundamental group is unchanged by the facet deletions, which occur in dimension $n - 1 \ge 5$; cellular approximation applies to the common two-skeleton. Hence F_m^- is simply connected. The homology form of Whitehead's theorem [3, Corollary 4.33], applied to its map to a point, proves contractibility. □

Rectangles and restricted bands.

Lemma 7. *The closed Manhattan Rips complex at radius k on a nonempty finite integer rectangle is contractible.*

Proof. By [1, Corollary 4.3], this complex is the nerve of the open Manhattan balls of radius $(k+1)/2$ centered at its grid points. The balls have convex intersections and cover the whole rectangular region: each point is within Manhattan distance one of a grid point. Coordinate clamping to the rectangle decreases distance to every grid center. The straight-line homotopy to the clamped point stays in each ball containing the starting point. Thus the union deforms onto the rectangle, and the nerve theorem [2, Theorem 10.7] proves the claim. □

For coordinate sets I, J, write $X|_{I \times J}$ for the induced restriction of a complex X.

Lemma 8. *If I, J are each a nonempty cyclic interval of length at most k or the whole circle, and at least one is a proper interval, then $L^-|_{I \times J}$ is contractible.*

Proof. Cover it by the restrictions of the punctured bands. A nonempty band restricted to a proper interval in its circular coordinate is a cone. In its cross-polytope description, at least one vertex has its opposite absent: an interval of length less than k has no antipodal pair, while one of length k has interior vertices with antipodes outside. No selected facet lies in this restricted direction, since its parity projection has minimum cyclic span $n - 2 > k$. If a full circular direction is retained, a complete band in that direction is its punctured ball. A restriction to one row or column is a simplex. All nonempty intersections of distinct restricted bands remain simplices.

For a full coordinate the band-intersection graph is an n-cycle. For a proper interval it is a path: its vertices are the consecutive bands meeting that interval, and its edges correspond to their shared coordinates in the interval. This includes the two bands meeting a singleton. The cover nerve is the join of the two graphs. It has a path factor, so it is contractible. The subcomplex nerve theorem completes the proof. □

4. Comparison and the homotopy formula

We use the following finite-cover homology observation.

Lemma 9. *Suppose a finite simplicial complex X is covered by subcomplexes $X_1, \dots, X_s$, and $Y \subset X$. If for every nonempty index set $J \subset \{1, \dots, s\}$ the inclusion*

$$Y \cap \bigcap_{j \in J} X_j \longrightarrow \bigcap_{j \in J} X_j$$

induces an isomorphism on integral homology, then $Y \hookrightarrow X$ does so.

Proof. Induct on s. Separate X_s from the union of the preceding members. The induction hypothesis applies both to that union and to its intersection with X_s, covered by $X_i \cap X_s$, $i < s$. Naturality of the Mayer–Vietoris sequences for these subcomplex unions and the five lemma give the assertion in every degree; see [3, Section 2.2, pp. 149–150]. □

Proof of Theorem 1. By Lemma 5, the induced restrictions to

$$\{a, \dots, a+k\} \times \mathbb{Z}/n\mathbb{Z}, \qquad \mathbb{Z}/n\mathbb{Z} \times \{b, \dots, b+k\},$$

cover K^- simplex by simplex. Use the same restrictions for L^-.

Every intersection is a restriction to $I \times J$. Each coordinate set is the whole circle if unconstrained; otherwise it is empty, a connected cyclic interval of length at most k, or a pair of antipodal points. Indeed, two closed semicircles either meet in an interval or, when complementary, in their two endpoints. Intersecting a shorter interval with another semicircle cannot disconnect it, since their total lengths are less than a full circumference. Further cuts of an endpoint pair give a singleton or the empty set.

If one coordinate is an antipodal pair, every clique is contained in one of its two rows or columns, or is a pair whose other coordinate is constant. All these faces already lie in L, and none is deleted. Thus the restricted inclusion from L^- to K^- is the identity. The same is true for a singleton coordinate.

If both coordinates are proper connected intervals, the metric unwraps to an ordinary rectangle. No selected facet lies there because $n - 2 > k$. Lemmas 7 and 8 make both restrictions contractible. If exactly one coordinate is proper and the other is full, Proposition 6 makes the restriction of K^- contractible, and Lemma 8 does the same for L^-. Empty intersections occur on both sides.

Lemma 9 now gives

$$H_*(L^-; \mathbb{Z}) \cong H_*(K^-; \mathbb{Z}).$$

By Proposition 4, the only nonzero reduced group of K^- is $H_3(K^-; \mathbb{Z}) \cong \mathbb{Z}$.

The complex K^- is simply connected. Each Rips edge is homotopic, relative to its endpoints, to a shortest grid path of length at most k. Its path vertices form a retained simplex, having at most $k + 1 < n$ vertices. Unit-square triangulations are also retained. Square homotopies commute horizontal and vertical steps, so a closed grid loop reduces to powers of the two coordinate cycles. These cycles bound in the retained full row and full column simplices.

Hurewicz first gives $\pi_2(K^-) = 0$ and then represents a generator of $H_3(K^-; \mathbb{Z})$ by a map $S^3 \to K^-$. This map is an integral homology isomorphism, so [3, Theorem 4.32 and Corollary 4.33] gives $K^- \simeq S^3$. Lemma 3 restores the $2n = 4k$ spheres of dimension $n - 1 = 2k - 1$, proving the theorem. □

AI assistance

OpenAI Codex was used to explore and draft the proof, check the cited theorem statements, and prepare the manuscript. Independent human mathematical review is requested.

References

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